Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2006
PI Name:	Jackson, Kenneth A. Ph.D.		
Project Title:	Growth of Rod Eutectics		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85712	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/24/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1667		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The study will involve computer modeling as well as experimental investigation of a model system. The experimental study will use transparent eutectic analogs to model the growth of eutectics in semiconductor/metal alloy systems, in order to observe the development of the microstructures, and to complement the theoretical work. The study will include modeling of the growth of facet/facet and facet/non-facet eutectic systems using Monte Carlo computer simulation methods. These will be atomic level simulations, which will incorporate the interface faceting during growth by appropriate scaling of the parameters, which describe the alloy system. The objective of this research is to develop our understanding and control of the microstructures in rod eutectics in which one or both phases are faceted. The microstructure in these alloys develops primarily during crystallization. We will study the development of the microstructure for various compositions, and under various growth conditions and growth rates. This will lead to an improved detailed understanding of how the microstructures develop. The microstructures found in our model systems and computer simulations will be directly compared with the microstructures found, for example, in		

	semiconductor-metal systems, and will ultimately provide improved control over the development of those microstructures. The phenomenology of these alloys is rich, and the in-situ studies will help to define and control the conditions for the growth of uniform coupled rod eutectic microstructures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)